

Tech Stack Recommendation

PU PS 6 — Intelligent Meeting, Decision and Action Tracking System

1. Core Pipeline

Stage	Tool	Why
Speech-to-text	Whisper (faster-whisper / whisper.cpp) self-hosted, or Deepgram / AssemblyAI free tier	Whisper large-v3 handles multi-speaker meeting audio well; free and runs local on a decent GPU. Cloud APIs give speaker diarization out of the box if not building it yourself.
Speaker diarization	pyannote.audio	Needed to know who said what — critical for responsibility assignment. Pairs with Whisper output via timestamp alignment.
Decision / action-item extraction	LLM-based extraction (orchestrator + free model — Llama 3.1/3.3 70B via Groq, or Gemini Flash free tier)	Prompt classifies each transcript segment into discussion / decision / action_item / info with structured JSON output — directly answers the PS requirement to distinguish content types.
Structured output enforcement	Pydantic + instructor, or JSON schema constrained decoding	Forces the LLM to return clean {decision, owner, deadline, department} objects instead of prose that needs regex parsing.

2. Semantic Search & Knowledge Graph

Need	Tool	Why
Vector search over meeting history	ChromaDB (local, free) or Qdrant (scales better)	Embeds decisions/summaries so “searchable meeting knowledge” works with natural-language queries.
Embeddings	sentence-transformers (all-MiniLM-L6-v2) local, or Gemini/OpenAI embeddings cloud	Free, fast, good enough for meeting-scale text, no API cost.
Knowledge graph (optional)	NetworkX (lightweight, in-memory) or Neo4j Community (looks more serious)	Links decisions → people → departments → follow-up meetings. Good for department-wise dashboards and traceability. Don't over-engineer — NetworkX + a good schema is enough for academic scope.

3. Task Tracking & Escalation

Need	Tool	Why
Task database	PostgreSQL (or SQLite for smaller scope)	Stores action items with status, deadline, owner, escalation level.
Scheduled reminders / escalation	n8n	Cron-trigger workflow: check overdue items daily → notify owner → escalate to manager after N days. This is exactly what n8n is built for — reuse the existing setup.
Notifications	n8n → Email / Slack / WhatsApp node	Free tier covers this easily.

4. Dashboards

Need	Tool	Why
Analytics / dashboard	Streamlit, or Next.js + Recharts	Streamlit is fastest to build for a demo (Python, matches the extraction pipeline). Next.js if a more product-like look is wanted.
Meeting effectiveness metrics	Computed from the task DB — no new tool needed	% actions closed on time, average decision-to-action lag, department participation, via aggregation queries.

5. Suggested Architecture Flow

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Audio -> Whisper (transcribe) + pyannote (diarize)
-> LLM extraction (segment classify: discussion / decision / action / info)
-> Structured JSON (Pydantic validated)
-> PostgreSQL (tasks/decisions) + ChromaDB (semantic search index)
-> n8n (escalation cron + reminders)
-> Streamlit / Next.js dashboard (department views, analytics)
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6. Where to Spend Effort vs. Simplify for the Demo

Given hackathon / academic scope, don't build full diarization and a full knowledge graph if time is tight — that's a lot of surface area. Priority order for a working demo:

Whisper transcription — real, easy win

LLM extraction with structured output — this is the core “intelligence”, spend most time here

Task DB + basic dashboard — real, straightforward

Escalation — real, already a known tool

Diarization + knowledge graph — can be simplified or mocked for the demo; mention as future scope if time runs out